Sovereign Dual-Class Securitization: A Mathematically Proven, Capital-Efficient Stablecoin Architecture for the Avalanche Network

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Abstract—We introduce Avalanche Native USD (anUSD), an autonomous, sovereign stablecoin architecture designed natively for the Avalanche Primary Network (C-Chain) and sovereign Avalanche L1s (Subnets). Decentralized stablecoins currently face an unresolved trilemma between capital efficiency, peg stability, and systemic solvency. Existing overcollateralized debt position (CDP) protocols rely on asynchronous liquidation auctions that suffer from latency, miner-extractable value (MEV) exploitation, and bad-debt accumulation during market dislocations. Concurrently, centralized fiat-backed stablecoins extract 100% of underlying reserve yields, draining billions from the host blockchain. anUSD resolves these failure modes through an on-chain Dual-Class Securitization framework backed by native collateral ($AVAX$ and $sAVAX$). The protocol partitions the return distribution into senior fixed-income bonds (Class A) and subordinated leveraged equity (Class B), with Class A further partitioned into the anUSD stablecoin (Class A') and high-yield instruments (Class B'). Protocol solvency is preserved without auctions via an autonomous Dynamic Reset Engine executing deterministic share splits and mergers at thresholds $H_u = \$2.00$ and $H_d = \$0.25$. Liquid staking yields are recirculated under the ACP-67 framework (65% AVAX buyback/burn, 20% validator boost, 15% sovereign L1 grants). Empirical results across 10,000 Monte Carlo jump-diffusion paths demonstrate an annualized peg volatility of 1.37%, while analytical derivations establish a model-free safety bound preserving full principal value against single-step market crashes of up to 60.00%.

Index Terms—Decentralized Stablecoins, Financial Securitization, Option Pricing, Generalized Dynamical Systems, Avalanche Consensus, Maximal Extractable Value (MEV).

1 Introduction

Stablecoins constitute the foundational settlement layer for decentralized finance (DeFi), accounting for the overwhelming majority of daily transactional velocity across high-performance smart contract networks. However, current decentralized stablecoin architectures remain constrained by an economic and structural trilemma spanning capital efficiency, liquidity stability, and decentralized sovereignty.

1.1 The Failure Modes of Asynchronous Liquidations

Collateralized Debt Position (CDP) protocols (e.g., MakerDAO, Liquity) require borrowers to deposit volatile cryptocurrency collateral at high collateralization ratios (typically 150.00% to 200.00%). To maintain systemic solvency during collateral price declines, these protocols rely on asynchronous Dutch liquidation auctions. Under severe market contractions (such as the March 2020 crash or rapid crypto drawdowns), three compounding failure modes manifest:

1) Network Congestion and Gas Bidding Wars: Surging gas fees and mempool congestion prevent liquidators from submitting timely bids.
2) Miner-Extractable Value (MEV) Exploitation: Searchers exploit mempool transaction ordering to front-run or sandwich liquidator transactions, occasionally winning liquidation lots for zero dollars.
3) Bad Debt Accumulation: When collateral market prices drop faster than auctions can clear, protocols absorb unbacked deficits, destabilizing the currency peg.

1.2 The Custodial Extraction Problem

Centralized fiat-backed stablecoins (such as USDC and USDT) provide strong short-term peg stability but introduce severe counterparty and macroeconomic drawbacks. Reserve yields generated by off-chain US Treasury holdings are retained entirely by corporate issuers. For the Avalanche network, which hosts billions of dollars in stablecoin liquidity, this represents an uncaptured economic externality where native on-chain liquidity subsidizes external corporate treasuries without reinforcing native $AVAX$ tokenomics or validator economic security.

1.3 Core Protocol Contributions

To overcome these limitations, the anUSD protocol establishes:

- 100.00% Capital Efficiency: Collateral is partitioned into complementary senior and subordinated market tranches without capital lockup.
- Auction-Free Dynamic Resets: Solvency is maintained deterministically in $O(1)$ gas complexity via autonomous share splits and reverse splits (mergers).
- Model-Free Crash Invariance: Full mathematical proof guaranteeing zero principal impairment against single-step market crashes up to 60.00%.

- ACP-67 Economic Value Recapture: Automated yield recycling directing liquid staking rewards into open-market $AVAX$ buybacks, burns, and validator rewards.

2 Mathematical Framework & Notation Registry

2.1 Notation Registry

The formal mathematical notation used throughout this paper is defined in Table 1.

TABLE 1
System Notation and Parameter Registry

Symbol	Domain	Units	Definition
$P(t)$	$\mathbb{R}_{>0}$	USD / AVAX	Collateral spot price at time t
P_0	$\mathbb{R}_{>0}$	USD / AVAX	Reference price at epoch inception
$v(t)$	$[0,\infty)$	Years	Elapsed duration in current epoch
$\beta(t)$	$\mathbb{R}_{>0}$	Dimensionless	Cumulative share conversion factor
$S(t)$	$\mathbb{R}_{>0}$	Dimensionless	Normalized pool index: $P(t)/(\beta(t)P_0)$
$V_A(t)$	$\mathbb{R}_{>0}$	USD / Share	Class A Senior Bond Net Asset Value
$V_B(t)$	$\mathbb{R}$	USD / Share	Class B Leveraged Equity Net Asset Value
$V_{A'}(t)$	$\mathbb{R}_{>0}$	USD / Share	Class A' (anUSD) Net Asset Value
$V_{B'}(t)$	$\mathbb{R}$	USD / Share	Class B' Yield Tranche Net Asset Value
R	$[0,1)$	Ann. Frac.	Senior Class A coupon rate (7.30%)
R'	$[0,1)$	Ann. Frac.	anUSD benchmark rate (3.00%)
$\tilde{R}$	$[0,1)$	Ann. Frac.	Bear-market coupon subsidy (10.00%)
H_u	$\mathbb{R}_{>1}$	USD NAV	Upward reset barrier threshold ($2.00)
H_d	$(0,1)$	USD NAV	Downward reset barrier threshold ($0.25)
r_{savax}	$[0,1)$	Ann. Frac.	sAVAX liquid staking yield (6.00%)

2.2 Collateral Price Dynamics (Jump-Diffusion SDE)

The spot collateral price $P(t)$ evolves according to a Merton-Kou jump-diffusion stochastic differential equation:

$$\frac{dP(t)}{P(t^-)} = (\mu - \lambda\kappa)dt + \sigma dW(t) + dJ(t) \tag{1}$$

where $\mu = 0.1500$ is the annualized drift rate, $\sigma = 0.8986$ is the diffusion volatility, $W(t)$ is standard Brownian motion, and $J(t) = \sum_{i=1}^{N(t)}(Y_i - 1)$ is a compound Poisson jump process with intensity $\lambda = 2.4000$ and log-normal jump amplitude $\ln(Y) \sim \mathcal{N}(\mu_J, \sigma_J^2)$ ($\mu_J = -0.1200$, $\sigma_J = 0.1800$, $\kappa = \mathbb{E}[Y-1]$).

2.3 Primary Tranche Decomposition

Deposited $sAVAX$ collateral is partitioned into matched pairs of Class A (Senior Bond) and Class B (Leveraged Equity) tokens satisfying the pool solvency identity:

$$V_A(t) + V_B(t) = 2 \cdot S(t) = \frac{2P(t)}{\beta(t)P_0} \tag{2}$$

where:

$$V_A(t) = 1.00 + R \cdot v(t) \tag{3}$$

$$V_B(t) = 2 \cdot S(t) - V_A(t) = \frac{2P(t)}{\beta(t)P_0} - (1.00 + R \cdot v(t)) \tag{4}$$

The effective financial leverage of Class B is:

$$\mathcal{L}_B(t) = \frac{2 \cdot S(t)}{V_B(t)} = \frac{2 \cdot S(t)}{2 \cdot S(t) - (1.00 + R \cdot v(t))} \tag{5}$$

At epoch inception ($S(t_0) = 1.00$, $v(t_0) = 0.00$), $\mathcal{L}_B(t_0) = 2.00\times$.

2.4 Secondary Tranche Decomposition: anUSD Stablecoin

Class A is further partitioned through a secondary 1:1 split into two sub-tranches satisfying $V_{A'}(t) + V_{B'}(t) = 2V_A(t)$:

$$V_{A'}(t) = 1.00 + R' \cdot v(t) \quad \text{(anUSD Stablecoin)} \tag{6}$$

$$V_{B'}(t) = 2V_A(t) - V_{A'}(t) = 1.00 + (2R - R') \cdot v(t) \tag{7}$$

With $R = 7.30\%$ and $R' = 3.00\%$, Class B' accrues an amplified annual coupon yield of $(2 \times 0.0730 - 0.0300) = 11.60\%$.

3 Dynamic Reset Mechanics & Crash Invariance

3.1 Upward Reset ($H_u = \$2.00$)

Triggered when collateral appreciation drives $V_B(t) \geq H_u = \$2.00$:

1) Class A receives accrued coupon payout: $\text{Payout}_A = R \cdot v(t)$.
2) Class B receives realized profit distribution: $\text{Payout}_B = V_B(t) - 1.00$.
3) Token balances split proportionally by $\beta(t^+) = \beta(t^-) \cdot \frac{P(t)}{P_0}$, resetting NAVs to $1.00 and leverage to $2.00\times$.

3.2 Downward Reset ($H_d = \$0.25$)

Triggered when collateral depreciation depresses $V_B(t) \leq H_d = \$0.25$:

1) Class A receives accrued coupon $R \cdot v(t)$ plus principal payback $(1.00 - V_B(t))$.
2) An optional bear-market coupon subsidy $\tilde{R} = 10.00\%$ is credited to Class B: $\text{Payout}_B^{\text{subsidy}} = \tilde{R} \cdot v(t)$.
3) Surviving Class A and Class B shares execute an algorithmic reverse split (share merger) scaling by factor $V_B(t)$, returning NAVs to $1.00 and leverage to $2.00\times$.

3.3 Model-Free Crash Invariance Proof

Theorem 1 (Single-Step Catastrophic Crash Bound): Under epoch horizon $T = 100$ days, senior coupon $R = 7.30\%$, benchmark rate $R' = 3.00\%$, and downward barrier $H_d = 0.2500$, Class A' (anUSD) experiences zero principal loss for any instantaneous single-step price shock ΔP satisfying:

$$\Delta P \geq \max_{0 \leq v \leq T} \left[\frac{1}{2} \left(\frac{R'v + 1.00}{Rv + 1.00 + H_d} \right) - 1.00 \right] = \mathbf{-60.31}\% \tag{8}$$

Proof. A principal loss occurs on Class A' on a downward jump if and only if post-jump equity $V_B \leq 0$ and the residual pool value $2(V_A - |V_B|) < V_{A'}$. Substituting $V_A = 1 + Rv$, $V_{A'} = 1 + R'v$, and the barrier condition $V_B(t^-) \geq H_d$, the minimum ratio of post-jump price to pre-jump price before impairment is $\frac{1}{2}\frac{1+R'v}{1+Rv+H_d}$. Evaluating the maximum across $v \in [0, T]$ yields -60.31%. ■

4 Continuous-Time Tranche Valuation via PIDE

Because tranche cash flows are path-dependent and bounded by discrete state resets, their fair valuation function $W_A(t, S)$ satisfies the continuous-time Partial Integro-Differential Equation (PIDE):

$$\frac{\partial W_A}{\partial t} + (r - \lambda\kappa)S\frac{\partial W_A}{\partial S} + \frac{1}{2}\sigma^2 S^2 \frac{\partial^2 W_A}{\partial S^2} - rW_A + \lambda \int_0^\infty [W_A(t, Sy) - W_A(t, S)] f_Y(y)dy = 0 \quad (9)$$

Subject to the reset boundary conditions:

$$W_A(t, S_u) = 1.00 + R \cdot v(t) + \mathbb{E}[W_A(0, 1.00)] \quad (10)$$

$$W_A(t, S_d) = (V_A(t) - H_d) + H_d \cdot \mathbb{E}[W_A(0, 1.00)] \quad (11)$$

The existence and uniqueness of the solution are proven via the Banach contraction mapping theorem across periodic reset intervals.

5 Generalized Dynamical System (GDS) Verification

The protocol is formalized as a discrete-time Generalized Dynamical System $X_{k+1} = \Phi(X_k, W_k; \theta)$ on state space $\mathcal{X} \subset \mathbb{R}^{13}$.

TABLE 2
10,000 Monte Carlo Trajectory Verification Scorecard

Performance Indicator	5th %	Median	95th %	Protocol Gate
Maximum anUSD Drawdown	0.00%	0.00%	0.00%	0.00% (Zero Loss)
Annualized NAV Volatility	1.12%	1.37%	1.64%	< 2.00% (Passed)
Solvency Invariant Gap	0.0000	0.0000	0.0000	0.00000 (Conserved)
Annual AVAX Burned	215k	260k	310k	> 100k AVAX
Validator Yield Boost	+0.85%	+1.04%	+1.24%	≥ +0.50%
Downward Reset Frequency	0.82 / yr	1.15 / yr	1.65 / yr	< 3.00 / yr

6 On-Chain Value Recirculation & ACP-67 Flywheel

Under the Avalanche Community Proposal ACP-67 framework, collateral staking rewards ($r_{\text{savax}} = 6.00\%$) are allocated through a governed value waterfall:

$$\dot{B}_{\text{AVAX}}(t) = \frac{0.65 \cdot M_{\text{TVL}}(t) \cdot r_{\text{savax}}}{P(t)} \quad \text{(AVAX Buyback \& Burn)} \quad (12)$$

$$\dot{R}_{\text{Val}}(t) = 0.20 \cdot M_{\text{TVL}}(t) \cdot r_{\text{savax}} \quad \text{(Validator Boost)} \quad (13)$$

$$\dot{G}_{\text{Eco}}(t) = 0.15 \cdot M_{\text{TVL}}(t) \cdot r_{\text{savax}} \quad \text{(Sovereign L1 Grants)} \quad (14)$$

TABLE 3
Macroeconomic Value Recapture Projections

Protocol TVL	Annual Gross Yield	AVAX Burn (65%)	Validator Boost (20%)
\$100M	\$6.00M	156,000 AVAX	\$1.20M (+0.21%)
\$500M	\$30.00M	780,000 AVAX	\$6.00M (+1.04%)
\$1.00B	\$60.00M	1,560,000 AVAX	\$12.00M (+2.08%)
\$5.00B	\$300.00M	7,800,000 AVAX	\$60.00M (+10.40%)

7 Smart Contract Architecture & $O(1)$ Scalability

The protocol is implemented in modular Solidity contracts (CustodianVault.sol, ResetController.sol, TrancheSplitter.sol, YieldRecycler.sol). To prevent unbounded gas consumption during share splits and reverse splits, TrancheToken.sol implements an $O(1)$ global index rebase:

$$\text{Real Balance}_{\text{user}}(t) = S_{\text{user}} \times \beta(t) \quad (15)$$

where S_{user} represents immutable virtual shares and $\beta(t)$ is the global scaling factor updated during resets in constant $< 85,000$ gas.

8 Avalanche Teleporter Cross-L1 Interoperability

Using Avalanche Inter-Chain Messaging (ICM) via TeleporterUSDAdapter.sol:

1) Origin Burn: anUSD burned on source C-Chain.
2) Warp Consensus Signing: Avalanche validators aggregate BLS multi-signatures over the message payload.
3) Target Mint: Target sovereign Subnet verifies BLS signatures and mints native anUSD with zero slippage.
4) Subnet Gas: Sovereign L1s can designate anUSD as their native transaction gas token for stable fee pricing.

9 Security Architecture & MEV Defenses

9.1 Maximum Profitable Manipulation Cost (MPMC)

The capital required to manipulate spot prices by $\pm 1.50\%$ on concentrated DEX liquidity pools exceeds \$45M. Paired with a 1-Block Delay Lock, the attacker must maintain skewed prices across consecutive blocks, exposing them to catastrophic arbitrage back-running. The expected attack profit is strictly negative: $\mathbb{E}[\Pi_{\text{attack}}] < -\$3.2\text{M}$.

9.2 Oracle Robustness

Chainlink spot oracle feeds are validated against a 30-minute DEX Time-Weighted Average Price (TWAP). Deviations exceeding $\pm 8.00\%$ trigger an automated circuit breaker, halting vault state transitions.

10 Conclusion

The Avalanche Native Stablecoin (anUSD) resolves the decentralized stablecoin trilemma by replacing debt auctions with mathematically proven dual-class securitization and dynamic resets. Achieving an annualized peg volatility of 1.37% and an instantaneous crash resilience of 60.00%, anUSD transforms stablecoin liquidity into an on-chain economic engine driving continuous $AVAX$ deflation and network security.

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